# Retention-constrained class expansion with reference-anchored candidate tables

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**Abstract**

A trained classifier contains discrimination that can support new-class recognition under an explicit original-class accuracy requirement. We develop reference-anchored candidate tables to make this expansion capacity measurable and usable. Fixed base references normalize learned, semantic and support-derived evidence without changing existing scores; a separate offset controls novel admission. Matched-retention frontiers quantify capacity, paired sequential comparisons quantify the retained classifier's value, and independent protection-risk calibration selects admissible thresholds. Experiments use three datasets, ten class partitions and two frozen CLIP encoders. On ImageNet-R, the ViT-L/14 frontier reaches 84.31% novel accuracy at 86.42% base accuracy. Retaining the classifier adds 0.71 and 0.90 percentage points over dense prototypes across five arrivals with ViT-L/14 and ViT-B/32. Independent calibration accepts expansion on all ten ImageNet-R partitions per encoder, yielding 77.64% and 45.30% novel accuracy while meeting every five-session test-retention target. Sparse calibration pools select an explicit base-only fallback. These results connect retained discrimination to useful

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expansion under stated protection requirements.
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## 1. Introduction

An expanding recognition system already has a classifier whose decision boundaries reflect task-specific supervision. New classes arrive with names and a few examples. The central question is how much new-class recognition that existing discrimination can support while maintaining a specified accuracy on the original classes. A second question follows across repeated arrivals: how much predictive value is gained by keeping the trained classifier rather than replacing it with prototypes or recalibrated semantic evidence?

We address these questions with *reference-anchored candidate tables*. The table retains learned base columns and appends semantic anchors and support prototypes for arriving classes. Each evidence branch is normalized against fixed base references, so adding candidates leaves existing scores unchanged. An independent admission offset controls competition between base and novel classes. This separates the discrimination already learned from the decision to admit new alternatives.

The contribution is the connection between retained discrimination and class-expansion capacity: a persistent score frame makes the original classifier reusable, a common base-accuracy constraint makes its admission value comparable, and paired arrival sequences measure its continuing benefit. Independent protection-risk calibration adds a confidence-qualified admission rule. On ImageNet-R, the gain over dense prototypes persists across two

frozen encoders; the independently calibrated tables admit novel classes on every evaluated partition. Complete comparisons also identify calibration budgets and datasets where the selected policy returns the base classifier.

## 2. Related work

Long-tailed recognition addresses unequal class supervision through balanced fitting and losses [1, 2, 3]. CLIP supplies semantic image–text evidence [4], while prompt adaptation [5, 6], prototypes [7, 8] and semantic caches [9] adapt pre-trained representations to labeled data. Our setting combines these evidence types according to their availability at class arrival.

Class-incremental and few-shot incremental learning study retention across successive tasks [10, 11]. SimpleCIL [12] and RanPAC [13] show the value of pre-trained prototypes and analytic readouts. KBS approaches develop multi-feature similarity supplementation [14], sharpness-aware prototype calibration [15], adaptive exploration [16], representation-enhanced analytic learning [17], and multimodal prompt fusion [18]. CLIP-guided discovery [19] additionally uses unlabeled novel data, whereas our arriving classes provide names and labeled support. FeNeC [20] supplies a recent clustered-representative readout for controlled frozen-feature comparisons.

BiMC [21] and HyCal [22] calibrate multimodal or heterogeneous recognition metrics. Calibrated seen/unseen recognition [23] studies the related operating trade-off, and Learn then Test [24] supplies risk-controlled threshold selection. We focus on the retained classifier as a source of admission capacity: its contribution is measured at a common base requirement, over successive arrivals, and under a separately calibrated protection policy.

## 3. From retained discrimination to novel admission

### *3.1. Task and evidence*

Base classes $\mathcal{B}$ have labeled, potentially imbalanced fitting data. Arriving classes $\mathcal{N}$ provide a name and $k$ labeled support images per class. Prediction selects from $\mathcal{B} \cup \mathcal{N}$ without a group oracle. A frozen encoder produces a normalized feature $v = f(x)$; normalized text anchors $t_c$ average eight fixed prompts. Learned base scores are $q_L(x, b) = v^\top w_b + b_b$, fitted by inverse-frequency-weighted ridge regression. Novel prototypes are normalized support means $p_n$, giving $q_P(x, n) = v^\top p_n$; semantic scores are $q_T(x, c) = v^\top t_c$.

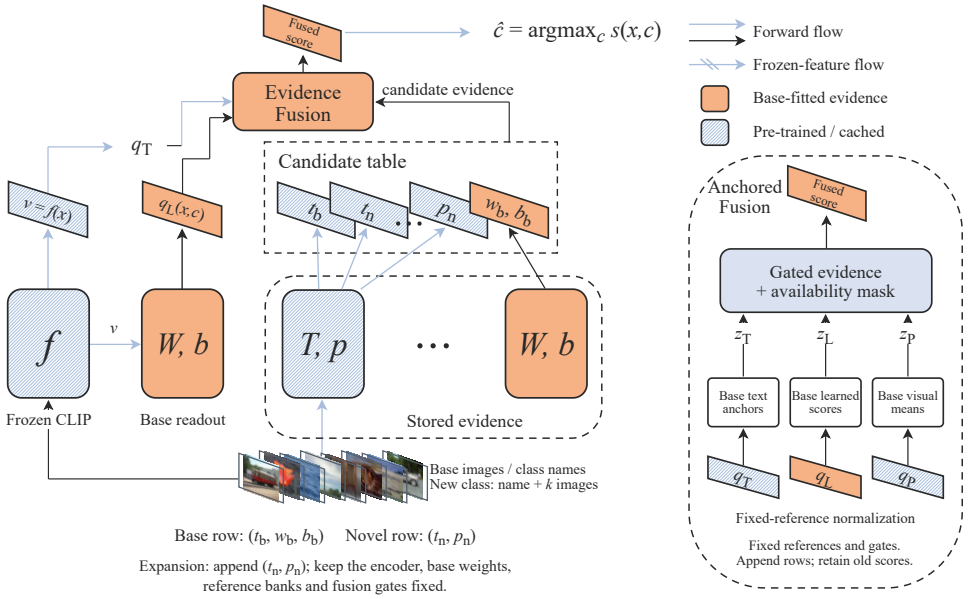



Figure 1: Reference-anchored candidate tables. Fixed base references set each branch's score frame; learned base columns remain stored as novel text anchors and support prototypes are appended. The separate novel offset controls admission. Orange marks base-fitted evidence; blue hatching marks pre-trained or cached evidence.

For each evidence source $e$, a fixed base reference bank $\mathcal{R}_e$ defines

$$\mu_e(x) = |\mathcal{R}_e|^{-1} \sum_{b \in \mathcal{R}_e} q_e(x, b), \tag{1}$$

$$\sigma_e^2(x) = |\mathcal{R}_e|^{-1} \sum_{b \in \mathcal{R}_e} (q_e(x, b) - \mu_e(x))^2. \tag{2}$$

We use base text anchors, learned columns and visual means as the text, learned and prototype references. Normalized evidence is

$$\widehat{q}_e(x, c) = \frac{q_e(x, c) - \mu_e(x)}{\max(\sigma_e(x), 10^{-6})}. \tag{3}$$

The statistics adapt to each query while remaining independent of the active novel vocabulary. Fusion uses the evidence available to each row:

$$s(x, b) = \widehat{q}_T(x, b) + a\widehat{q}_L(x, b), \qquad b \in \mathcal{B}, \tag{4}$$

$$s(x, n) = \widehat{q}_T(x, n) + u\widehat{q}_P(x, n) + d, \qquad n \in \mathcal{N}. \tag{5}$$

The coefficients $a, u$ combine evidence; $d$ controls novel admission (Fig. 1). Candidate-wise normalization instead changes the branch scales when new rows arrive, which can alter existing fused decisions. A numerical example appears in Supplementary Section K.

**Proposition 1.** *With fixed input, encoder, existing row evidence, reference banks, coefficients and tie-breaking, appending rows preserves every existing score and its within-vocabulary ordering.*

All terms of an existing row remain fixed. This structural property also holds for raw and fixed-temperature controls; their recognition comparisons isolate reference normalization. Appended rows can still win the expanded

prediction. Supplementary Section E gives the proof and prediction consequence. Each new row costs $O(D)$ storage and $O(kD)$ prototype construction after encoding; scoring $K$ classes costs $O(KD)$.

### *3.2. Calibration and expansion capacity*

A base-only pseudo-expansion episode splits base classes into pseudo-base and pseudo-novel groups. A classifier fitted on pseudo-base classes is combined with names and reserved support for pseudo-novel classes. Held-out base queries select $a, u, d$ by harmonic macro accuracy,

$$H = 2A_B A_N/(A_B + A_N), \tag{6}$$

with zero assigned when both accuracies vanish. The coefficients then remain fixed across actual arrivals. Actual novel query labels and test labels are excluded from this selection; grids and split seeds are in Supplementary Section K.

Let $A_{B,\mathrm{ref}}^{\mathrm{closed}}$ denote the retained table's closed-base macro accuracy. At retention fraction $r$, every readout receives the same absolute base target. Its empirical expansion capacity is

$$N_r = \max_d A_N(d) \quad \text{subject to} \quad A_B(d) \geq r A_{B,\mathrm{ref}}^{\mathrm{closed}}. \tag{7}$$

A test-label sweep characterizes this descriptive frontier; an operational threshold uses calibration data. The exact margin envelope is derived in Supplementary Section D. Reporting these two evaluations separately distinguishes attainable capacity from selected-policy behavior.

### *3.3. Independent protection-risk calibration*

Let $R$ indicate correct prediction by the retained closed-base reference and $C_d$ correct prediction by the expanded readout. Define

$$L_d = R(1 - C_d) + (1 - r)(1 - R). \tag{8}$$

Writing $D = \Pr(R = 1, C_d = 0)$ gives $\mathbb{E}L_d = D + (1 - r)(1 - A_{B,\mathrm{ref}})$. Thus $\mathbb{E}L_d \leq 1 - r$ implies $A_B(d) \geq rA_{B,\mathrm{ref}}$. This common sufficient criterion conservatively ignores replacement readouts' recovery of reference errors.

Fitting and tuning data fix the score models and ordered candidate thresholds. Independent, class-stratified calibration observations test thresholds from conservative to permissive using weighted bounded-loss p-values and fixed-sequence testing [24]. Testing stops at the first failure. Allocating error probability $\delta/5$ per arrival controls population protection across five prespecified sessions for each method, order and partition under independent class-conditional sampling. It does not guarantee arbitrary distribution shifts or every finite test outcome. If no finite threshold passes, the policy returns the reference classifier with zero novel admission. Supplementary Section J provides the bound, assumptions and full selection rule.

## 4. Experiments

### *4.1. Protocol and comparisons*

We evaluate CIFAR-100 [25], Oxford-IIIT Pets [26] and ImageNet-R [27], using frozen OpenAI CLIP ViT-L/14 and ViT-B/32 features, five-shot support, ten frequency-stratified class partitions and two five-session arrival orders. Base/novel counts are 40/40, 15/15 and 100/100; each session adds eight,

three or 20 classes. Base pools follow long-tailed sampling. ImageNet-R uses task-specific splits after removing byte-identical duplicates. Novelty is relative to task fitting, with CLIP's pre-training exposure retained. Detailed pools, preprocessing and seeds are in Supplementary Sections C, I and K.

Every comparator shares its encoder, partition and support. We compare fixed-gate classifier removal, independently recalibrated removal, dense anchored text–prototype fusion, summed/class-mean Tip-Adapter caches, and raw/fixed-temperature retained fusion. The latter uses a single base-validation scale per branch. All mixtures use base-only selection; FeNeC and earlier three-partition auxiliary studies are retained in Supplementary Sections A–H.

Orders are averaged within a partition. Means and standard deviations describe ten class assignments on fixed images. Paired differences use identical cells; constrained comparisons use jointly feasible sets. Conditional paired 95% bootstrap intervals resample partition means 20,000 times. Average incremental accuracy is the mean macro accuracy over five arrivals. These intervals characterize class-assignment variation, conditional on the fixed data and encoders.

### *4.2. Novel recognition at a shared base requirement*

At the descriptive 95% retention budget, the ImageNet-R L/14 table reaches 84.31% novel accuracy at 86.42% base accuracy. Figure 2 shows the complete budgets across encoders and datasets. A more discriminative base readout can tolerate greater novel competition while meeting the same target, turning retained discrimination into admission capacity.

Reference normalization also improves the sequential comparison against fixed-temperature fusion on ImageNet-R by 0.94 [0.74, 1.12] percentage points

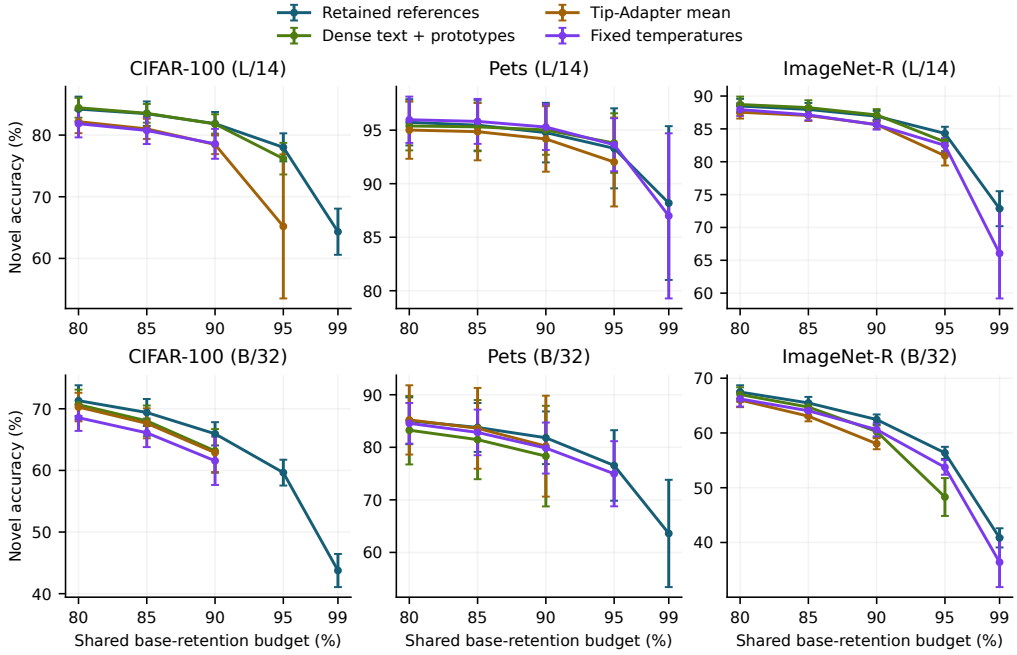



Figure 2: Descriptive matched-retention capacity. Readouts share the retained model's absolute base target within each partition/encoder. Error bars show partition standard deviations; plotted points require all ten partitions to be feasible. Offsets use test labels for this capacity analysis. Partial feasible sets and counts appear in Supplementary Section I.

with L/14. Both preserve existing scores, so this comparison measures the value of query-dependent branch scaling. Full normalization controls are in Supplementary Section I.

### *4.3. The classifier's value across arrivals*

Table 1 quantifies retained-minus-comparator average incremental accuracy with fixed base-only selected gates and offsets. On ImageNet-R, retaining the classifier adds 0.71 [0.46, 0.92] points over dense prototypes with L/14 and 0.90 [0.69, 1.11] with B/32. Corresponding gains over class-mean caches are 1.42 [1.15, 1.62] and 1.71 [1.44, 1.98] points. The result persists across both encoders and after calibration is restored to the classifier-free alternative.

CIFAR also shows gains over recalibrated removal with both encoders; its dense-prototype intervals span zero. Pets' average incremental intervals span zero for these comparators. The gain over prototype replacement is clearest on ImageNet-R.

### *4.4. Turning capacity into calibrated admission*

An empirical calibration threshold selects the largest offset satisfying $r$ times the reference's calibration accuracy. We separately evaluate whether test accuracy meets $r$ times its test reference. In the original L/14 pools, the retained 95% policy is validation-feasible in all ten partitions but meets the test target in 7/10 CIFAR, 1/10 Pets and 2/10 ImageNet-R partitions. Supplementary Section I reports all methods; Section H varies threshold-label budgets.

For independent risk calibration, every readout is refitted after repartitioning the original fitting/validation pool into approximately 40% fitting, 20%

Table 1: Paired retained-classifier gain in average incremental accuracy, percentage points [conditional paired 95% bootstrap interval]. Two orders are averaged within each of ten partitions. Full final-session differences are in Supplementary Section I.

| Encoder | Dataset | Comparator | Avg. gain [95% CI] |
|---|---|---|---|
| L/14 | CIFAR-100 | Recalibrated removal | 3.00 [2.23, 3.75] |
| L/14 | CIFAR-100 | Dense text + prototypes | 0.15 [-0.46, 0.64] |
| L/14 | CIFAR-100 | Tip-Adapter mean | 1.44 [0.68, 2.10] |
| L/14 | Pets | Recalibrated removal | -0.14 [-1.53, 0.86] |
| L/14 | Pets | Dense text + prototypes | -0.98 [-2.52, 0.22] |
| L/14 | Pets | Tip-Adapter mean | -0.06 [-1.76, 1.41] |
| L/14 | ImageNet-R | Recalibrated removal | 2.30 [1.95, 2.59] |
| L/14 | ImageNet-R | Dense text + prototypes | 0.71 [0.46, 0.92] |
| L/14 | ImageNet-R | Tip-Adapter mean | 1.42 [1.15, 1.62] |
| B/32 | CIFAR-100 | Recalibrated removal | 2.38 [1.44, 3.34] |
| B/32 | CIFAR-100 | Dense text + prototypes | 0.86 [-0.15, 1.76] |
| B/32 | CIFAR-100 | Tip-Adapter mean | 0.96 [-0.04, 1.84] |
| B/32 | Pets | Recalibrated removal | -0.44 [-1.64, 0.81] |
| B/32 | Pets | Dense text + prototypes | -0.39 [-1.85, 1.02] |
| B/32 | Pets | Tip-Adapter mean | -0.74 [-1.62, 0.12] |
| B/32 | ImageNet-R | Recalibrated removal | 3.07 [2.82, 3.34] |
| B/32 | ImageNet-R | Dense text + prototypes | 0.90 [0.69, 1.11] |
| B/32 | ImageNet-R | Tip-Adapter mean | 1.71 [1.44, 1.98] |

tuning and 40% calibration observations per class. Support and test pools remain separate. The comparison uses these newly fitted models throughout, with $r = 0.95$, $\delta = 0.05$ and 101 tuning-margin quantiles as candidate thresholds. The benchmarks have previously informed method development; this is a revised benchmark evaluation.

Table 2: Independent protection-risk calibration on ImageNet-R at 95% retention and 95% five-session confidence. Values are mean test percentages across ten partitions; Exp. counts finite accepted thresholds. Rejected expansion returns the shared reference with zero novel accuracy, included in the means. Complete datasets and confidence levels appear in Supplementary Section J.

| Encoder | Readout | Exp. | Test met | Base | Novel |
|---|---|---|---|---|---|
| L/14 | Retained | 10/10 | 10/10 | 89.73 | 77.64 |
| L/14 | Recalibrated removal | 0/10 | 10/10 | 91.54 | 0.00 |
| L/14 | Dense prototypes | 5/10 | 10/10 | 91.04 | 24.12 |
| L/14 | Raw fusion | 4/10 | 10/10 | 91.08 | 26.42 |
| L/14 | Fixed temperature | 10/10 | 10/10 | 89.76 | 65.69 |
| L/14 | Class-mean cache | 2/10 | 10/10 | 91.18 | 6.20 |
| B/32 | Retained | 10/10 | 10/10 | 73.66 | 45.30 |
| B/32 | Recalibrated removal | 0/10 | 10/10 | 75.01 | 0.00 |
| B/32 | Dense prototypes | 0/10 | 10/10 | 75.01 | 0.00 |
| B/32 | Raw fusion | 0/10 | 10/10 | 75.01 | 0.00 |
| B/32 | Fixed temperature | 7/10 | 10/10 | 74.24 | 27.28 |
| B/32 | Class-mean cache | 0/10 | 10/10 | 75.01 | 0.00 |

Retained tables accept expansion on all ten ImageNet-R partitions with both encoders (Table 2), reaching 77.64% novel accuracy with L/14 and 45.30% with B/32. All final test targets are met, as are all five targets along each of the 20 arrival paths per encoder. Under the common sufficient protection criterion, L/14 fixed-temperature fusion accepts ten partitions at 65.69% novel accuracy, dense prototypes five, and class-mean caches two. On CIFAR and Pets, sparse tail-class calibration budgets select base-only fallback in every retained-table partition with both encoders; these outcomes represent abstention from expansion, with zero novel recognition.

## 5. Conclusion

Candidate tables turn retained discrimination into reusable expansion capacity. Fixed references preserve the existing score frame, while an admission threshold exposes the trade-off between original-class protection and novel recognition. Matched-retention and paired sequential comparisons quantify this value; independent calibration turns it into useful protected admission on ImageNet-R with two encoders. The framework connects what an existing classifier has learned to how much new recognition it can support.

## CRediT authorship contribution statement

Junyi Zhang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing.

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## Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The experiments use public CIFAR-100, Oxford-IIIT Pets and ImageNet-R data and the publicly released CLIP ViT-L/14 and ViT-B/32 checkpoints. Code, source-hashed experimental records, prediction archives, manuscript sources, and table/figure generation scripts are available at `https://github.com/Cheerssom/Reference-Anchored-Candidate-Tables`. Dataset images, pretrained weights, and feature caches are obtained separately using the repository instructions.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI Codex was used to assist with literature discovery, manuscript drafting and revision, experiment and verification scripts, and LaTeX and table preparation. The author is responsible for the accuracy, interpretation and originality of the submitted work and for reviewing and approving the final manuscript and supporting materials.

## References


[1] Z. Liu, Z. Miao, X. Zhan, J. Wang, B. Gong, S. X. Yu, Large-Scale Long-Tailed Recognition in an Open World, arXiv preprint arXiv:1904.05160 (2019).
URL https://arxiv.org/abs/1904.05160

[2] K. Cao, C. Wei, A. Gaidon, N. Arechiga, T. Ma, Learning Imbalanced Datasets with Label-Distribution-Aware Margin Loss, arXiv preprint arXiv:1906.07413 (2019).
URL https://arxiv.org/abs/1906.07413

[3] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, S. Belongie, Class-Balanced Loss Based on Effective Number of Samples, arXiv preprint arXiv:1901.05555 (2019).
URL https://arxiv.org/abs/1901.05555

[4] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, I. Sutskever, Learning Transferable Visual Models From Natural Language Supervision, in: International Conference on Machine Learning, 2021, pp. 8748–8763.
URL https://proceedings.mlr.press/v139/radford21a.html

[5] K. Zhou, J. Yang, C. C. Loy, Z. Liu, Learning to Prompt for Vision-Language Models, International Journal of Computer Vision 130 (9) (2022) 2337–2348. doi:10.1007/s11263-022-01653-1.
URL https://link.springer.com/article/10.1007/s11263-022-01653-1

[6] J.-X. Shi, T. Wei, Z. Zhou, J.-J. Shao, X.-Y. Han, Y.-F. Li, Long-Tail Learning with Foundation Model: Heavy Fine-Tuning Hurts, in: International Conference on Machine Learning, 2024, pp. 45014–45039.
URL https://proceedings.mlr.press/v235/shi24g.html

[7] J. Snell, K. Swersky, R. S. Zemel, Prototypical Networks for Few-shot Learning, arXiv preprint arXiv:1703.05175 (2017).
URL https://arxiv.org/abs/1703.05175

[8] Y. Wang, W.-L. Chao, K. Q. Weinberger, L. van der Maaten, SimpleShot: Revisiting Nearest-Neighbor Classification for Few-Shot Learning, arXiv preprint arXiv:1911.04623 (2019).
URL https://arxiv.org/abs/1911.04623

[9] R. Zhang, W. Zhang, R. Fang, P. Gao, K. Li, J. Dai, Y. Qiao, H. Li, Tip-Adapter: Training-free Adaption of CLIP for Few-shot Classification, arXiv preprint arXiv:2207.09519 (2022).
URL https://arxiv.org/abs/2207.09519

[10] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, C. H. Lampert, iCaRL: Incremental Classifier and Representation Learning, arXiv preprint arXiv:1611.07725 (2016).
URL https://arxiv.org/abs/1611.07725

[11] X. Tao, X. Hong, X. Chang, S. Dong, X. Wei, Y. Gong, Few-Shot Class-Incremental Learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 12183–12192.

URL https://openaccess.thecvf.com/content_CVPR_2020/html/Tao_Few-Shot_Class-Incremental_Learning_CVPR_2020_paper.html

[12] D.-W. Zhou, Z.-W. Cai, H.-J. Ye, D.-C. Zhan, Z. Liu, Revisiting Class-Incremental Learning with Pre-Trained Models: Generalizability and Adaptivity are All You Need, International Journal of Computer Vision (2024).
URL https://github.com/LAMDA-CL/RevisitingCIL

[13] M. D. McDonnell, D. Gong, A. Parvaneh, E. Abbasnejad, A. van den Hengel, RanPAC: Random Projections and Pre-trained Models for Continual Learning, in: Advances in Neural Information Processing Systems, Vol. 36, 2023, pp. 12022–12053. doi:10.52202/075280-0526.
URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/2793dc35e14003dd367684d93d236847-Abstract-Conference.html

[14] X. Xu, S. Niu, Z. Wang, W. Guo, L. Jing, H. Yang, Multi-feature space similarity supplement for few-shot class incremental learning, Knowledge-Based Systems 265 (2023) 110394. doi:10.1016/j.knosys.2023.110394.

[15] R. Chen, X.-Y. Jing, F. Wu, H. Chen, Sharpness-aware gradient guidance for few-shot class-incremental learning, Knowledge-Based Systems 299 (2024) 112030. doi:10.1016/j.knosys.2024.112030.

[16] C. Han, Z. Gu, C. Xu, Z. Cui, Adaptive exploration for few-shot

incremental learning, Knowledge-Based Systems 330 (2025) 114496. doi:10.1016/j.knosys.2025.114496.

[17] R. He, D. Fang, Y. Chen, K. Tong, C. Chen, Y. Wang, L.-p. Chau, H. Zhuang, REAL: Representation enhanced analytic learning for exemplar-free class-incremental learning, Knowledge-Based Systems 332 (2026) 114901. doi:10.1016/j.knosys.2025.114901.

[18] H. Fang, H. Cai, K. Mao, J. Zhao, X. Xiao, Few-shot class-incremental learning based on prompt guidance and multimodal fusion, Knowledge-Based Systems 342 (2026) 115862. doi:10.1016/j.knosys.2026.115862.

[19] Q. Yan, Y. Yang, Y. Dai, X. Zhang, K. Wiltos, M. Woźniak, W. Dong, Y. Zhang, CLIP-guided continual novel class discovery, Knowledge-Based Systems 310 (2025) 112920. doi:10.1016/j.knosys.2024.112920.

[20] K. Książek, H. Jastrzębski, K. Pniaczek, B. Trojan, M. Karp, J. Tabor, FeNeC: Enhancing continual learning via feature clustering with neighbor- or logit-based classification, Knowledge-Based Systems 338 (2026) 115479. doi:10.1016/j.knosys.2026.115479.

[21] Y. Chen, T. Ding, L. Wang, J. Huo, Y. Gao, W. Li, Enhancing few-shot class-incremental learning via training-free bi-level modality calibration, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025, pp. 9881–9890.
URL https://openaccess.thecvf.com/content/CVPR2025/html/Ch

en_Enhancing_Few-Shot_Class-Incremental_Learning_via_Training-Free_Bi-Level_Modality_Calibration_CVPR_2025_paper.html

[22] E. Lee, M. Kim, J. Kwon, Y. Lee, J. Kim, S. Jang, Y. Kim, Hycal: A training-free prototype calibration method for cross-discipline few-shot class-incremental learning, arXiv:2604.15678v2 (2026).
URL https://arxiv.org/abs/2604.15678

[23] W.-L. Chao, S. Changpinyo, B. Gong, F. Sha, An Empirical Study and Analysis of Generalized Zero-Shot Learning for Object Recognition in the Wild, arXiv preprint arXiv:1605.04253 (2016).
URL https://arxiv.org/abs/1605.04253

[24] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, L. Lei, Learn then test: Calibrating predictive algorithms to achieve risk control, arXiv preprint arXiv:2110.01052 (2021). doi:10.48550/arXiv.2110.01052.

[25] A. Krizhevsky, Learning multiple layers of features from tiny images, Tech. rep., University of Toronto (2009).
URL https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf

[26] O. M. Parkhi, A. Vedaldi, A. Zisserman, C. V. Jawahar, Cats and dogs, in: IEEE Conference on Computer Vision and Pattern Recognition, 2012, pp. 3498–3505. doi:10.1109/CVPR.2012.6248092.
URL https://www.robots.ox.ac.uk/~vgg/publications/2012/parkhi12a/

[27] D. Hendrycks, S. Basart, N. Mu, S. Kadavath, F. Wang, E. Dorundo, R. Desai, T. Zhu, S. Parajuli, M. Guo, D. Song, J. Steinhardt, J. Gilmer, The Many Faces of Robustness: A Critical Analysis of Out-of-Distribution Generalization, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 8340–8349.
URL https://openaccess.thecvf.com/content/ICCV2021/html/Hendrycks_The_Many_Faces_of_Robustness_A_Critical_Analysis_of_Out-of-Distribution_ICCV_2021_paper.html